ML: MULTIPLE CHOICE QUESTIONS - 4

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1. **What is a neural network inspired by?**

   A. Human digestive system
   B. Human brain
   C. Computer processors
   D. Internet structure

2. **What is the basic unit of a neural network?**

   A. Node (Neuron)
   B. Layer
   C. Weight
   D. Bias

3. **Which of the following is NOT a type of neural network?**

   A. CNN
   B. RNN
   C. SVM
   D. ANN

4. **What does ANN stand for?**

   A. Artificial Neural Network
   B. Automated Neural Node
   C. Artificial Numeric Network
   D. Advanced Neural Network

5. **What is the role of weights in a neural network?**

   A. Store data
   B. Adjust input importance
   C. Define output labels
   D. Control learning rate

6. **What is a bias in a neural network?**
   A. Error term
   B. Constant added to output
   C. Activation function D. Input data

7. **Which function introduces non-linearity?**

   A. Loss function
   B. Activation function
   C. Cost function
   D. Gradient function

8. **Which is a common activation function?**

   A. Sigmoid
   B. Linear Regression
   C. K-means
   D. PCA

9. **What is ReLU?**

   A. Regression Unit
   B. Rectified Linear Unit
   C. Random Linear Utility
   D. Recursive Learning Unit

10. **What does CNN stand for?**

    A. Central Neural Network
    B. Convolutional Neural Network
    C. Continuous Neural Network D. Computational Neural Network

11. **CNNs are mainly used for:**

    A. Text processing

B. Image processing
C. Audio processing
D. Numerical sorting

12. What does RNN stand for?

A. Recursive Neural Network
B. Recurrent Neural Network
C. Random Neural Network D. Regression Neural Network

13. RNNs are suitable for:

A. Static images
B. Sequential data
C. Random data
D. Tabular data

14. What is backpropagation?

A. Forward data passing
B. Error correction method
C. Data normalization
D. Feature selection

15. What is a loss function?

A. Measures model error
B. Activates neurons
C. Stores weights
D. Optimizes input

16. Which is a loss function?

A. Mean Squared Error
B. Sigmoid
C. ReLU

D. Dropout

17. What is overfitting?

A. Model performs well on all data
B. Model performs poorly on training data
C. Model memorizes training data but fails on new data
D. Model ignores data

18. What is dropout?

A. Removing data
B. Ignoring some neurons during training
C. Decreasing learning rate D. Increasing data size

19. What is an epoch?

A. One full pass through dataset
B. One neuron update
C. One layer
D. One output

20. What is batch size?

A. Number of layers
B. Number of neurons
C. Number of samples per update D. Number of outputs

21. What does gradient descent do?

A. Maximizes loss B. Minimizes loss
C. Removes data
D. Adds noise

22. What is learning rate?

A. Speed of training
B. Number of neurons

C. Number of layers
D. Size of dataset

23. What is a hidden layer?

A. Input layer
B. Output layer
C. Intermediate layer
D. Final result

24. What is softmax used for?

A. Regression
B. Classification probabilities
C. Data cleaning
D. Feature scaling

25. Which library is commonly used for neural networks?

A. NumPy
B. TensorFlow
C. Pandas
D. Matplotlib